

# ConstantPad2d#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.padding.ConstantPad2d.html>

## Description:

Pads the input tensor boundaries with a constant value. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses (padding\_left\text{padding\\_left}padding\_left, padding\_right\text{padding\\_right}padding\_right, padding\_top\text{padding\\_top}padding\_top, padding\_bottom\text{padding\\_bottom}padding\_bottom) Input: (N,C,Hin,Win)(N, C, H\_{in}, W\_{in})(N,C,Hin,Win) or (C,Hin,Win)(C, H\_{in}, W\_{in})(C,Hin,Win). Output: (N,C,Hout,Wout)(N, C, H\_{out}, W\_{out})(N,C,Hout,Wout) or (C,Hout,Wout)(C, H\_{out}, W\_{out})(C,Hout,Wout), where

## Function Signature

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```
class torch.nn.modules.padding.ConstantPad2d(padding, value)  
[source]#
```

Parameters

Shape:

## Code Examples

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```
>>> m = nn.ConstantPad2d(2, 3.5)
>>> input = torch.randn(1, 2, 2)
>>> input
tensor([[[ 1.6585,  0.4320],
          [-0.8701, -0.4649]]])
>>> m(input)
tensor([[[[ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  1.6585,  0.4320,  3.5000,  3.5000],
          [ 3.5000,  3.5000, -0.8701, -0.4649,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000]]]])
>>> # using different paddings for different sides
>>> m = nn.ConstantPad2d((3, 0, 2, 1), 3.5)
>>> m(input)
tensor([[[[ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  1.6585,  0.4320],
          [ 3.5000,  3.5000,  3.5000, -0.8701, -0.4649],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000]]]])
```

# Detailed Documentation

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## Documentation

Pads the input tensor boundaries with a constant value.

For N-dimensional padding, use `torch.nn.functional.pad()`.

`padding` (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses  $(padding\_left, padding\_right, padding\_top, padding\_bottom)$

Input:  $(N, C, H_{in}, W_{in})$  or  $(C, H_{in}, W_{in})$

Output:  $(N, C, H_{out}, W_{out})$  or  $(C, H_{out}, W_{out})$ , where

$H_{out} = H_{in} + padding\_top + padding\_bottom$

$W_{out} = W_{in} + padding\_left + padding\_right$